Notes: Brav, Cain & Zytnick (JFE, 2022)

Retail shareholder participation in the proxy process: Monitoring, engagement, and voting

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1 Big picture

- **Question.** How do retail shareholders participate in corporate proxy voting, and do their turnout and voting decisions reflect meaningful monitoring rather than mechanical passivity?
- **Data contribution.** The paper uses a nearly comprehensive proprietary Broadridge dataset of U.S. retail shareholder voting records from 2015–2017.
- **Turnout finding.** Retail turnout is low on average but rises with account stake, account value, potential benefits from winning, and easier voting access.
- **Monitoring finding.** Conditional on voting, retail shareholders punish poorly performing management teams.
- **Cost finding.** Notice-and-Access delivery rules can sharply reduce turnout among investors whose preferred voting method is not available.
- **Retail versus institutional finding.** Retail voting differs from institutional voting, especially in sensitivity to ISS recommendations.
- **Exit and voice.** Retail shareholders are more likely to continue holding firms after voting with management.

2 Data and measurement

2.1 Retail voting data

The core dataset comes from Broadridge Financial Solutions and covers annual and special meetings from 2015 through 2017. Each account-meeting observation includes whether the account voted, how it voted on each proposal, the number of shares held at the record date, and zip-code information. Broadridge anonymizes accounts but allows tracking across firms and years.

2.2 Proposal and firm data

The authors merge Broadridge retail votes with ISS Voting Analytics, SharkRepellent, CRSP, Compustat, Form N-PX mutual fund votes, Thomson Reuters ownership data, and demographic variables from public sources.

2.3 Main variables

Account stake size α is the account's shares divided by the firm's shares outstanding. *Cast* is whether the account submits a ballot. *For* is the share of votes in favor. *WithMGMT* equals one when the account votes with management's recommendation.

3 Models and empirical specifications

3.1 Rational-choice turnout

$$U = P \times B - C + D.$$

P is pivotality, B the benefit from winning, C voting cost, and D consumption or expressive benefit. The paper maps ownership stakes and firm benefits into $P \times B$, voting-method restrictions into C, and nonzero turnout by tiny accounts into D.

3.2 Turnout regression

$$Cast_{amct} = \beta_0 + \beta_1 \log(\alpha_{amct}) + \beta_2 X_{amct} + FE + \varepsilon_{amct}.$$

This regression tests whether turnout rises with ownership stake and other proxies for benefits, costs, and investor engagement.

3.3 Information-material cost shock

$$Cast_{act} = \delta_0 - \delta_{1A} SwitchHCtoN_c \times Post_{ct} \times Default_{a0} + \delta_{1B} SwitchNtoHC_c \times Post_{ct} \times Default_{a0} + FE + \varepsilon_{act}.$$

This triple-difference design compares Default accounts at switching firms before and after the firm changes delivery method.

3.4 Voting with management

$$WithMGMT_{apmct} = \beta_0 + \beta_1 \log(\alpha_{amct}) + \beta_2 X_{apmct} + FE + \varepsilon_{apmct}.$$

This specification studies vote direction conditional on turnout.

4 Identification and empirical design

The paper combines descriptive evidence, rich fixed effects, counterfactual vote simulations, and a quasi-experimental delivery-method design. Turnout regressions compare accounts within the same meeting, the same account-year, and sometimes the same account-firm relationship. The Notice-and-Access design compares Default and non-Default accounts around firm delivery-method switches. The key limitation is that many voting-choice tests remain observational, though rich fixed effects greatly reduce confounding.

5 Tables and figures

5.1 Table 1: Retail investor characteristics and Figure 1: Ownership characteristics by account value and firm size

Read: Voting rate rises from about 3% in the smallest account-value quintile to about 15% in the largest. Small firms have high retail ownership percentages, while large firms have many more retail accounts.

Detailed interpretation: Retail importance has two dimensions: ownership concentration in small firms and numerosity in large firms.

Economic message: Retail investors are heterogeneous; larger accounts are more engaged and more likely to vote.

Table 1
Retail investor characteristics.
This table reports information on retail investors covered in the retail dataset. Retail characteristics are generated as follows: First, for each firm meeting, we use each account's holdings on the record date as a "snapshot" of that account's yearly holdings in the firm. We keep only one meeting of each firm per year. Second, for each account, we aggregate the holdings in the portfolio at the account-year level. Number of firms in portfolio is defined as the number of firms in a given year for which the account holds shares on the firm's record date. Account value is defined as the sum of an account's individual firm stake values, where individual stake values are calculated as the product of the number of shares in the firm held by the account and the price of the stock at the end of the record date month, as provided by CRSP. Dividend yield is defined as the difference between the firm's buy-and-hold return with dividends and without dividends. The account-year-level composite dividend yield is calculated as the account's dividends received summed over the firms held by that account divided by the account's total portfolio value. Market abnormal return for an account is calculated as the buy-and-hold abnormal return, using the CRSP value-weighted index return as a benchmark, on the securities in the account, assuming the account held all securities for the past year. Firm purchase rate and sale rate are the portion of portfolio firms that have been added or removed in the past year, respectively. To evaluate characteristics of the home area of the accounts in the sample, we obtain adjusted gross income data at the zip-code level from the IRS website. Zip-code mean AGI refers to the mean-adjusted gross income in the account's zip code. Voting rate is defined as the number of ballots cast divided by number of voting opportunities. Panel A includes summary statistics by year. In Panel B, we first average each account value over its years in the data and then sort accounts into quintiles by account value. We then report the average of each of the characteristics described in Panel A.

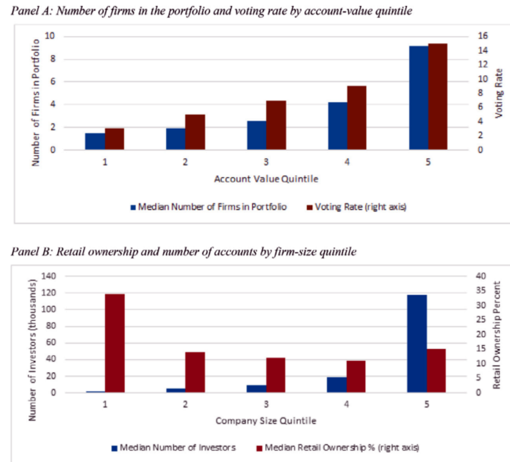
Panel A: Retail investor characteristics by year

	2015			2016			2017		
	Avg.	Med.	Stdev.	Avg.	Med.	Stdev.	Avg.	Med.	Stdev.
Num. of firms in portfolio	4.01	2.00	6.94	4.16	2.00	7.18	4.23	2.00	7.71
Account value	126,740	13,804	7,968,903	122,556	12,979	6,870,664	132,087	13,717	8,136,010
Dividend yield	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02
Market abnormal return	0.00	0.00	0.23	0.00	0.02	0.28	0.02	−0.03	0.33
Zip code mean AGI	102,502	76,596	87,867	105,404	79,198	89,372	104,148	79,925	83,580

Panel B: Average retail investor characteristics by account value

	Account value quintile				
	Smallest	2	3	4	Largest
Num. of firms in portfolio	1.47	1.88	2.54	4.20	9.16
Account value	588	4,077	13,131	39,814	649,064
Dividend yield	0.01	0.02	0.02	0.02	0.02
Market abnormal return	−0.03	0.00	0.02	0.02	0.02
Firm purchase rate	0.43	0.40	0.38	0.36	0.33
Firm sale rate	0.34	0.36	0.36	0.35	0.33
Zip code income	89,326	96,830	101,238	106,746	123,510
Voting rate	0.03	0.05	0.07	0.09	0.15

(a) Table 1



acteristics by account value and firm size.

(b) Figure 1

5.2 Table 2: Retail investor ownership characteristics and Table 3: Retail voting by meeting

Read: Domestic retail ownership averages roughly one-quarter of shares. Retail voters cast ballots representing about 32% of retail shares but only about 11% of retail accounts.

Detailed interpretation: Larger retail holders vote disproportionately, so share-weighted turnout exceeds account-weighted turnout.

Economic message: Retail shareholders own a lot collectively but face a severe account-level participation problem.

Table 2
Retail investor ownership characteristics.
This table reports information on ownership characteristics by retail shareholders. The sample is limited to proposals in the retail dataset that are matched with data from ISS Voting Analytics and CRSP. Panel A provides information on the number of investors and aggregate retail ownership for the full sample and across firm-size quintiles. Panel B provides information on the distribution of individual retail shareholders' equity stakes relative to the company's shares outstanding. For each firm-size quintile and for the full sample, we determine the average retail stake size, as well as the 25th, 50th, and 75th percentiles. Firm size is calculated as the product of CRSP variables csho and prc, and quintiles are determined using the NYSE size breakpoints from Ken French's website. "# Investors" refers to the number of retail investors in the sample, in thousands, who own shares in the firm. "Retail Ownership" is the percentage of outstanding shares of the firm held by domestic retail investors in the sample.

Panel A: Number of accounts and aggregate ownership

	2015				2016				2017			
	# Investors (thousands)		Retail ownership (%)		# Investors (thousands)		Retail ownership (%)		# Investors (thousands)		Retail ownership (%)	
Firm-size quintile:	Avg.	Median	Avg.	Median	Avg.	Median	Avg.	Median	Avg.	Median	Avg.	Median
Smallest	4	2	40	34	4	2	39	34	5	2	35	33
2	8	4	18	14	10	5	19	15	10	5	17	14
3	16	9	15	12	16	9	15	12	17	9	15	11
4	31	19	14	12	30	18	13	11	34	21	14	11
Largest	267	110	16	15	286	118	16	15	297	125	16	14
Full Sample	35	5	28	20	38	5	27	19	39	5	25	20

Panel B: Distribution of retail stake as a fraction of outstanding shares (in millionths)

	2015				2016				2017			
Firm-size quintile:	Avg.	25th	Median	75th	Avg.	25th	Median	75th	Avg.	25th	Median	75th
Smallest	84.17	1.10	5.50	22.49	74.71	0.94	5.09	22.06	72.54	0.53	3.40	16.53
2	21.10	0.68	2.51	7.80	18.00	0.29	1.59	5.96	15.93	0.26	1.38	5.27
3	10.03	0.35	1.32	3.90	9.27	0.26	1.04	3.37	8.35	0.20	0.98	3.26
4	5.48	0.20	0.69	1.89	3.34	0.17	0.61	1.75	5.56	0.16	0.55	1.60
Largest	0.61	0.03	0.08	0.26	0.58	0.02	0.08	0.26	0.53	0.02	0.07	0.23
Full Sample	6.27	0.03	0.13	0.58	5.42	0.03	0.13	0.54	6.11	0.03	0.12	0.52

Retail voting by meeting.
This table reports voting results at the ballot level. % Cast is the proportion of ballots cast as a fraction of the number of shares outstanding. % Voting only with management refers to ballots that entirely match management recommendations. % at least one against management refers to ballots with at least one vote that deviates from management's recommendation. The columns with header "Retail votes" are at the shareholder-vote level, and the columns with header "Retail account" are at the retail-account level, where each account is weighted equally.

	Retail votes			Retail accounts		
	% Cast	% Shares voting only with mgmt.	% At least one against mgmt.	% Cast	% accounts voting only with mgmt.	% At least one against mgmt.
All meetings	32	76	24	11	59	41
Annual meeting	32	76	24	11	58	42
Special meeting	38	79	21	15	74	26

by retail accounts, which is more reflective of the small pands on Table 3 by conditioning on the proposal types

5.3 Table 4: Retail voting and meeting proposals

Read: Retail voters support management proposals strongly but support shareholder proposals less than the overall electorate. Account-weighted and share-weighted retail results differ.

Detailed interpretation: Retail voters are distinct from institutions and from aggregate vote totals; they are not simply passive pro-management voters.

Table 4
Retail voting and meeting proposals.
This table reports information on retail voting limiting the sample to retail dataset proposals that are matched with data from ISS Voting Analytics and CRSP. Each entry represents the average of all firm votes in the category. "All votes" contains the overall voting results from ISS Voting Analytics, with corrections from SharkRepellent and CRSP, as described in Appendix A.3. "Retail Votes" contains domestic retail voting results. "Retail accounts" weighs domestic retail voting results at the account level. "Cast (%)" refers to the sum of the number of votes cast for and against divided by the number of potential votes as reported by ISS Voting Analytics. For and against votes exclude say-on-pay frequency votes and certain director votes for which the only retail voting data are on the number of votes cast. "For (%)" is the number of votes for divided by the number of votes cast. Panel A shows voting sorted by the identity of the sponsor, management, or shareholder. Panel B shows voting by sponsor and firm-size quintile. Panel C shows retail voting by proposal categories. Panel D shows voting sorted by sponsor and management and ISS recommendations.

Panel A: Retail voting by proposal sponsor

	All votes		Retail votes		Retail accounts	
	Cast (%)	For (%)	Cast (%)	For (%)	Cast (%)	For (%)
All	79	93	31	91	11	87
Management	79	95	31	93	11	89
Shareholder	75	30	28	18	11	29

Panel B: Retail voting by firm-size quintile

	All votes		Retail votes		Retail accounts	
	Cast (%)	For (%)	Cast (%)	For (%)	Cast (%)	For (%)
Management sponsored:						
Size quintile:						
Smallest	73	93	36	90	12	84
2	83	95	31	94	11	88
3	83	96	29	94	11	89
4	83	96	28	95	11	91
Largest	78	97	27	95	11	92
Shareholder sponsored:						
Size quintile:						
Smallest	70	45	43	39	12	46
2	81	47	35	26	10	40
3	82	38	29	22	12	33
4	79	36	28	22	11	32
Largest	74	27	27	15	11	26

Panel C: Retail voting by proposal category

	All votes		Retail votes		Retail accounts	
	Cast (%)	For (%)	Cast (%)	For (%)	Cast (%)	For (%)
Management:						
Elect director	78	97	29	95	11	93
Financial statements/Auditor	87	99	32	97	11	95
Governance - board and shareholder rights	77	94	33	92	12	88
Governance - compensation	74	90	32	87	11	76
Governance - other	77	91	40	90	14	84
Major transactions - issuance, buyback, distribution, stock split, or conversion	72	89	32	83	11	74
Major transactions - M&A	77	98	46	94	18	91
Other	78	82	34	89	12	87
Shareholder:						
Environmental	73	23	26	13	12	24
Social	74	19	27	15	11	27
Governance	77	38	29	21	11	31

Panel D: Retail voting by management and ISS recommendations

	All votes		Retail votes		Retail accounts	
	Cast (%)	For (%)	Cast (%)	For (%)	Cast (%)	For (%)
Management-sponsored:						
Management For & ISS For	79	97	30	94	11	89
Management For & ISS Against	72	76	34	87	10	80
Shareholder-sponsored:						
Management Against & ISS For	76	36	28	17	11	28
Management Against & ISS Against	73	8	26	14	12	25

Economic message: Retail shareholders form a distinct governance constituency.

5.4 Table 5: Impact of retail voting

Read: Removing or changing retail participation and preferences can flip close proposal outcomes. Changing retail preferences has effects comparable to changing Big Three preferences in close contests.

Detailed interpretation: The counterfactuals are mechanical outcome sensitivity exercises, not welfare calculations.

Economic message: Retail voice can matter in close corporate elections.

5.5 Table 6: Retail shareholder decision to cast a ballot and Figure 2: Relation between voter participation and ownership

Read: Turnout rises with ownership stake α and is concave in stake size. Even tiny-stake accounts sometimes vote.

Detailed interpretation: Financial incentives matter, but expressive or consumption benefits are also needed to explain nonzero turnout by very small shareholders.

Economic message: Retail voting is incentive-sensitive but not purely instrumental.

Table 5
Impact of retail voting.
This table describes changes in voting outcomes under hypothetical changes in the decision to vote, changes in retail ownership, and the voting preferences of certain groups of shareholders. Panel A provides the number of proposals whose outcome would change if a voting group's participation were set to zero. The sample consists only of proposals for which the voting base is the number of votes cast rather than the number of outstanding shares. We exclude routine proposals including auditor ratification and meeting adjournments, as well as director elections. We remove seven proposals for which the pass/fail outcome does not accord with the voting totals and rule for passage as reported by ISS Voting Analytics. Each row in Panel A designates a voting group whose participation is set to zero in the hypothetical. Columns (3) (and (4) reflect the number of proposals flipped under the hypothetical, and columns (5), (6), and (7) provide the number of proposals whose final percentage counts move by 5%, 10%, and 20%, respectively. Panel B provides the number of proposals whose outcome would change if ownership were shifted between retail and non-retail shareholders. We use the same sample as in Panel A and further remove two meetings in which the calculated retail ownership exceeds the number of outstanding shares from CRSP. We then change retail ownership by 17.9%, which is the standard deviation of retail ownership of all firms in the sample. Firms are sorted into quintiles of retail ownership, and we ask how an increase (decrease) in ownership for firms in the bottom, second, and third (third, fourth, and largest size) quintile affects vote outcomes. We report the consequences of these ownership changes separately for management and shareholder proposals. In Panel C, we hold fixed observed shareholder participation and report the number of proposals whose voting outcome would change if a voting group's preferences were altered. The two voting groups whose preferences we alter are those of retail shareholders, in the middle two columns, and the Big Three institutional investors, BlackRock, Vanguard, and State Street, in the right two columns. Voting choices are altered to the voting choice of the group described in the row header. To ensure a consistent comparison across the two voting groups, the number of votes we alter for a proposal is limited to the minimum of the number of retail votes and the number of Big Three votes. The sample in Panel C consists of the proposals in Panel A whose final overall number of votes in favor was between 4/5 and 6/5 of the number of votes required to pass. That is, for a standard proposal that would pass by a majority of cast ballots, Panel C limits to proposals that received 40% to 60% in favor. In all panels, columns (1) and (2) ("# passing proposals" and "# failing proposals") refer to the actual number of passing and failing proposals in each of the panel's samples. In Panel C, columns (3) and (4) reflect the number of proposals whose outcome is changed under the hypothetical that retail voters alter their voting preferences, and columns (5) and (6) reflect the number of proposals with changed outcomes under the hypothetical that the Big Three voters alter their voting preferences. In all panels, retail votes come from Broadridge and are limited to domestic retail shareholders, overall vote totals come from ISS's Voting Analytics dataset, and mutual fund votes come from a merge of Form N-PX, CRSP Mutual Funds, and Thomson Reuters S12 as described in Online Appendix E. In our counts of Big Three votes, we only include votes from N-PX for which we can match the fund to an ownership count for that firm from Form 13-F.

Panel A: Consequences due to shocks to retail participation

	Actual count		Change if group participation goes to zero				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Group whose participation goes to zero:	# passing proposals	# failing proposals	# passing proposals flipped to fail	# failed proposals flipped to pass	# of 5% movers	# of 10% movers	# of 20% movers
Retail voters	10,800	1,331	110	39	1,043	426	117
Big Three	10,800	1,331	46	62	368	26	3
All non-retail shareholders	10,800	1,331	385	159	7,430	4,758	2,000

Panel B: Consequences due to shocks to ownership structure

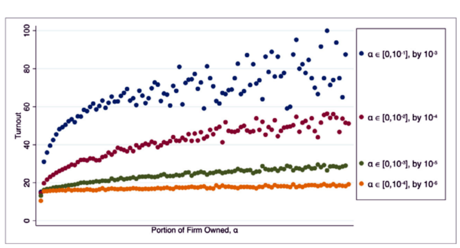
	Actual count		Change due to shocks to retail ownership				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Retail ownership quintile whose ownership is either increased or decreased:	# passing proposals	# failing proposals	# passing proposals flipped to fail	# failed proposals flipped to pass	# of 5% movers	# of 10% movers	# of 20% movers
Management proposals:							
Bottom quintile, + stdev.	2,155	26	0	3	16	4	0
Second quintile, + stdev.	2,102	54	1	9	29	1	0
Third quintile, + stdev.	2,003	35	0	9	48	5	0
Third quintile, - stdev.	2,003	35	10	0	32	1	0
Fourth quintile, - stdev.	2,027	30	16	0	65	6	1
Top quintile, - stdev.	2,300	28	21	1	223	20	0
Shareholder proposals:							
Bottom quintile, + stdev.	44	203	3	0	3	0	0
Second quintile, + stdev.	43	228	5	0	9	1	0
Third quintile, + stdev.	58	333	8	0	22	1	0
Third quintile, - stdev.	58	333	0	7	14	0	0
Fourth quintile, - stdev.	46	322	0	8	29	5	0
Top quintile, - stdev.	16	72	0	3	26	9	0

(*continued on next page*)

Table 5
(*continued*)

Panel C: Consequences due to shocks to retail voting preferences

	Actual count		Retail voters alter vote		Big Three voters alter vote	
	(1)	(2)	(3)	(4)	(5)	(6)
	# passing proposals	# failing proposals	# passing proposals flipped to fail	# failed proposals flipped to pass	# passing proposals flipped to fail	# failed proposals flipped to pass
Management proposals:						
Group voting decisions to adopt:						
Retail voters	235	87	0	0	14	24
Big Three	235	87	45	5	0	0
All non-retail shareholders	235	87	47	0	31	14
All in favor	235	87	0	17	0	32
All opposed	235	87	111	0	82	0
Shareholder proposals:						
Group voting decisions to adopt:						
Retail voters	60	159	0	0	14	6
Big Three	60	159	0	12	0	0
All non-retail shareholders	60	159	0	23	4	21
All in favor	60	159	0	58	0	59
All opposed	60	159	1	0	17	0



en voter participation and ownership.
relation between retail voter turnout and ownership of the firm. We plot a binned scatterplot of turnout on stake size

(a) Figure 2

Table 6
Retail shareholder decision to cast a ballot.
This table provides regression results describing retail shareholder turnout decisions. The dependent variable is equal to 1 if the account casts a ballot, and 0 otherwise, multiplied by 100. α is defined as the account's number of shares held divided by the firm's number of shares outstanding as of the record date month, from CRSP. Log market equity is the log of market equity, computed as price time shares outstanding from CRSP as of the record date month. Yearly abnormal return refers to the firm buy-and-hold return for the period 13 months to 1 month prior to the record date minus the value-weighted market return from CRSP. The dividend indicator is a binary variable equal to 1 if the difference in the firm's return with dividends and without dividends (*ret* and *retx* from CRSP, respectively) is positive. Tobin's q is book value plus market equity minus book equity, divided by book value. ROA, return on assets, is EBITDA divided by total assets. Special meeting is a binary variable equal to 1 for special meetings. Institutional ownership is equal to the number of shares owned by institutions divided by the shares outstanding, in the year prior to the meeting, both from Thomson Reuters. SRI on ballot is a binary variable equal to 1 if any proposals at the meeting are shareholder environmental or social proposals. Shareholder governance on ballot is a binary variable equal to 1 if any proposal at the meeting is a shareholder governance proposal. Log (Number of proposals on ballot) is the log of the number of proposals on the ballot. Log account value is the log of the total value of the account in the calendar year, defined as the sum across all firms held by the account of the product of share price and number of shares owned. 2016 county presidential turnout is the number of county residents who cast ballots in the 2016 U.S. presidential election, obtained from CQ Voting and Elections, divided by the number of adult citizens from the Census Bureau. Log Zip code AGI is the average adjusted gross income in the prior calendar year in the account's zip code. Fraction over 65 is the fraction of zip-code residents above age 65, from the Census, defined as $(DPSF0010015 + DPSF0010016 + DPSF0010017 + DPSF0010018 + DPSF0010019)/DPSF0010001)$. Density is the population divided by land area in square meters ($DPSF0010001/AREALAND$). Fraction with bachelors and fraction with post-bachelors are zip-code-level five-year averages from the U.S. Census as of 2017. Fraction in Finance/Insurance is equal to the number of employed workers in Finance/Insurance divided by all-industries employment, both at the zip-code level, from the Bureau of Labor Statistics. In Panel A, columns (1)–(2) use meeting fixed effects, columns (3)–(4) use industry and account-year fixed effects, columns (5)–(6) use meeting and account-year fixed effects, and column (7) uses meeting, account-year, and account-firm fixed effects. In Panel B, columns (1)–(3) use no fixed effects, column (4) uses industry and account-year fixed effects, and columns (5) and (6) use meeting fixed effects. In Panel B, columns (1)–(3), the reference category is accounts with ownership less than or equal to 10^{-9}. In addition, column (2) is limited to meetings in which no proposal comes within 30 percentage points of a different outcome, and column (3) is limited to accounts with account stake values of under \$100. Industry fixed effects use Fama French industry categories. In Panel A and columns 4–6 of Panel B, all right-hand-side variables are demeaned, so that the intercept reflects the turnout of an observation with average levels of each covariate. Observations are weighted by the inverse of the number of meetings for the account-year, so that each account-year is weighted equally. Standard errors clustered at the account and meeting level are in parentheses. Number of clusters refers to the number of distinct meetings. *, **, and *** represent significance at the 0.05, 0.01, and 0.001 levels, respectively.

Panel A: Retail shareholder turnout decisions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log(α)	1.751***	0.890	0.250***	0.601***	0.585***	0.250	0.573***
	(0.055)	(0.501)	(0.020)	(0.025)	(0.026)	(0.250)	(0.106)
Log(ME)				0.470***			
				(0.036)			
Yearly abnormal return				0.187			
				(0.221)			
Dividend indicator				0.139			
				(0.097)			
Tobin's q				−0.119***			
				(0.032)			
ROA				−0.845**			
				(0.279)			
Special meeting				4.871***			
				(0.427)			
Institutional ownership				−1.047***			
				(0.162)			
Log(α) × Log(ME)		0.052*				0.021	
		(0.024)				(0.011)	
Log(α) × Tobin's q		−0.204***				−0.031*	
		(0.027)				(0.013)	
Log(α) × ROA		0.384				−0.445***	
		(0.253)				(0.129)	
Log(α) × Special Meeting		1.535***				1.681***	
		(0.251)				(0.220)	
Log(α) × Inst. owner.		−0.074				−0.209**	
		(0.203)				(0.067)	
Intercept	7.865***	7.921***	9.422***	9.422***	9.429***	9.652***	10.460***
	(0.020)	(0.045)	(0.063)	(0.053)	(0.001)	(0.055)	(0.011)
Industry FE			Yes	Yes			
Meeting FE	Yes	Yes			Yes	Yes	Yes
Account-Year FE			Yes	Yes	Yes	Yes	Yes
Account-Firm FE							Yes
N	6,894,950	6,753,702	6,183,205	6,047,147	6,183,191	6,047,134	4,440,020
Number of clusters	8,264	7,874	8,271	7,880	8,260	7,870	7,644
R^2	0.04	0.04	0.79	0.80	0.80	0.80	0.88

(*continued on next page*)

(b) Table 6, Panel A

Table 6
(*continued*)

Panel B: Retail shareholder turnout decisions and consumption benefits of voting

	(1) Full sample	(2) No close proposals	(3) Stake less than \$100	(4) Full sample	(5) Full sample	(6) Full sample
$\alpha > 10^{-6}$	2.448***	3.134***	1.157***			
	(0.205)	(0.242)	(0.332)			
$\alpha > 10^{-7}$	2.722***	2.396***	0.408			
	(0.126)	(0.178)	(0.250)			
$\alpha > 10^{-8}$	3.007***	2.702***	1.328*			
	(0.189)	(0.419)	(0.542)			
$\alpha > 10^{-9}$	0.745***	0.469	−0.344			
	(0.205)	(0.460)	(0.535)			
SRI on ballot				0.133		
				(0.082)		
Shareholder governance on ballot				0.253**		
				(0.086)		
Log(Number of proposals on ballot)				−0.063		
				(0.103)		
Log(α)				0.587***	1.596***	1.611***
				(0.024)	(0.056)	(0.055)
Log(ME)				0.421***		
				(0.039)		
Institutional ownership				−1.237***		
				(0.157)		
Special meeting				4.874***		
				(0.417)		
Log account value					0.371***	0.369***
					(0.009)	(0.010)
2016 county presidential turnout					1.611***	
					(0.332)	
Log zip code income						−1.184***
						(0.097)
Fraction over 65						14.117***
						(0.537)
Density						−0.000***
						(0.000)
Fraction with bachelors						−0.430
						(0.518)
Fraction with post-bachelors						−0.427
						(0.801)
Fraction in Finance/Insurance						20.872***
						(2.514)
Intercept	2.677***	2.297***	2.397***	9.351***	7.992***	8.569***
	(0.127)	(0.179)	(0.119)	(0.034)	(0.025)	(0.054)
Industry FE				Yes		
Meeting FE					Yes	Yes
Account-Year FE				Yes		
N	6,894,960	2,757,938	276,723	6,056,453	6,456,515	6,352,277
Number of clusters	8,274	6,094	7,556	7,910	8,215	8,214
R^2	0.01	0.01	0.00	0.80	0.04	0.04

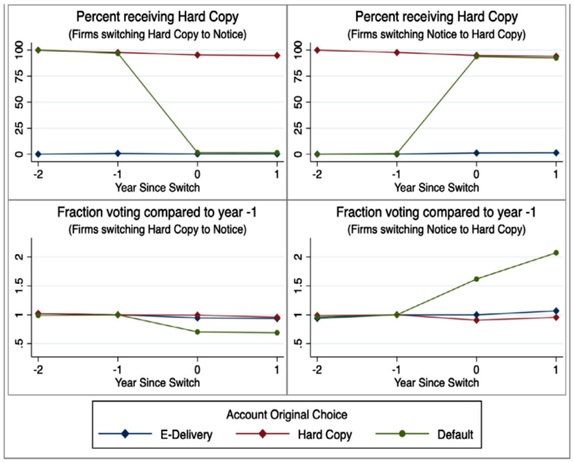
Table 6, Panel B

5.6 Figure 3 and Table 7: Information materials and turnout

Read: Switching from Hard Copy to Notice reduces turnout among Default accounts; switching in the opposite direction increases turnout.

Detailed interpretation: The triple-difference design turns delivery-method changes into a cost shock to voting access.

Economic message: Proxy-plumbing frictions materially affect retail shareholder voice.



(a) Figure 3

Table 7
Effect of information materials on turnout.
This table reports regression results describing how the availability of information materials shapes shareholder turnout decisions. The sample is limited to annual meetings held at firms that switch delivery methods a single time over the sample period 2015–2017 (along with 200 additional randomly selected firms that did not switch delivery methods). We further restrict the sample to firms and accounts that appeared in the data in 2015. The first column presents the estimation of the likelihood of receiving Hard Copy Materials (multiplied by 100) on the triple interaction of (i) whether the firm switches its information materials choice in the sample period (separated by the direction of the switch), (ii) whether the meeting in question is post-switch, and (iii) the proportion of firms in the account's portfolio in 2015 for which the account chose Default information materials. The second column presents the estimation of the likelihood of casting a ballot (multiplied by 100) on the triple-interaction terms. The third column presents the estimation of the likelihood of casting a ballot (multiplied by 100) on the receipt of Hard Copy Materials by scaling column (2) by column (1), following Duflo (2001). All regressions include account-firm, account-year, and firm-year fixed effects. Standard errors clustered at the account and meeting level are in parentheses. *, **, and *** represent significance at the 0.05, 0.01, and 0.001 levels, respectively.

	(1)	(2)	(3)
	Hard Copy Materials × 100	Cast × 100	Cast × 100
$\text{SwitchHCtoN}_c \times \text{Post}_{ct} \times \text{Default}_{a0}$	−91.657***	−2.554***	
	(0.807)	(0.254)	
$\text{SwitchNtoHC}_c \times \text{Post}_{ct} \times \text{Default}_{a0}$	89.934***	4.202***	
	(1.398)	(0.849)	
$\text{Hard Copy Materials}_{act}$			3.248***
			(0.317)
Account-Firm FE	Yes	Yes	Yes
Account-Year FE	Yes	Yes	Yes
Firm-Year FE	Yes	Yes	Yes
N	1,406,262	1,406,262	1,406,262
Number of clusters	306	306	306
R^2	0.98	0.94	0.94

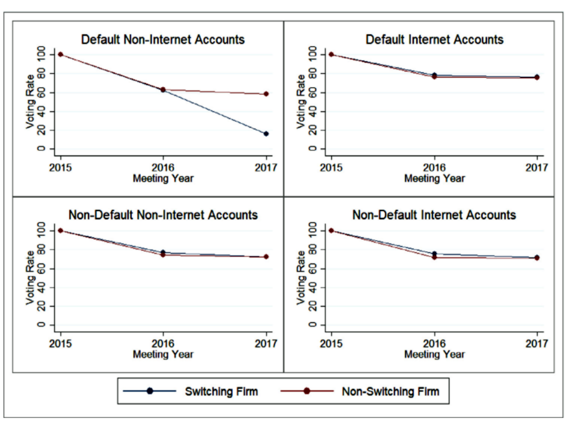
(b) Table 7

5.7 Figure 4 and Table 8: Available voting methods and turnout

Read: The largest turnout decline occurs for Default non-internet accounts when the firm changes delivery method.

Detailed interpretation: The relevant cost is not merely information receipt but access to a preferred voting method.

Economic message: Increasing digital delivery can reduce participation if it removes channels used by some investors.



voting method, delivery-method selection, and turnout by Default and non-Default accounts.

(a) Figure 4

Table 8
Effect of available voting methods on turnout.
This table reports regression results documenting how the availability of voting methods impacts shareholder turnout decisions. The sample is limited to annual meetings held at firms that switch delivery methods a single time over the sample period 2015–2017 (along with 200 additional randomly selected firms that did not switch delivery methods). The sample is further limited to firms that selected Hard Copy delivery methods in 2015 and to accounts that appeared in the data in 2015. The left two columns are limited to accounts that voted in 2015, with voting in 2016 as the dependent variable, and exclude switching firms that did not switch in 2016. The right two columns are limited to accounts that voted in 2016, with voting in 2017 as the dependent variable, and exclude switching firms that did not switch in 2017. The first and third columns contain accounts that selected Default; the second and fourth columns contain accounts that did not select Default. The right-hand-side variables include (i) whether the firm switched delivery methods, (ii) whether the account, when voting the previous year, voted by internet, and (iii) their interaction. Standard errors clustered at the account and meeting level are in parentheses. *, **, and *** represent significance at the 0.05, 0.01, and 0.001 levels, respectively.

	2016		2017	
	Default shareholders	Non-default shareholders	Default shareholders	Non-default shareholders
Switching firm × Did not vote by internet	−44.830***	1.157	−46.582***	−1.839
	(2.888)	(2.259)	(1.853)	(2.634)
Switching firm	−0.873	1.169	0.725	0.850
	(1.567)	(1.612)	(1.763)	(1.696)
Did not vote by internet	−13.772***	2.913	−15.373***	3.159
	(2.603)	(1.750)	(1.119)	(2.160)
Intercept	76.587***	71.350***	78.134***	74.559***
	(1.267)	(0.802)	(1.172)	(0.953)
N	15,818	55,176	25,230	90,742
Number of clusters	114	114	122	122
R^2	0.221	0.002	0.291	0.001

(b) Table 8

5.8 Table 9: Retail shareholder voting decisions

Read: Retail shareholders are less supportive of management at poorly performing firms and less sensitive to ISS than institutions.

Detailed interpretation: Conditional on voting, retail investors appear to monitor management rather than merely rubber-stamp proposals.

Economic message: Retail voting communicates dissatisfaction with weak management.

Table 9

Retail shareholder voting decisions.

This table reports evidence on account-level voting decisions with observations at the account-proposal level. The dependent variable is a binary variable that equals 1 if the account voted in line with management's recommendation, and 0 if it voted against, multiplied by 100. The analysis is limited to account proposals in which the account voted on the proposal and excludes routine proposals (auditor ratification and meeting adjournment). α is defined as the account's number of shares divided by the firm's number of shares outstanding on the record date month, from CRSP. Log market equity is the log of market equity computed as price time shares outstanding from CRSP, as of the record date month. Yearly abnormal return refers to the firm buy-and-hold return for the period 13 months to 1 month prior to the record date minus the value-weighted market return from CRSP. The dividend indicator is a binary variable equal to 1 if the difference in the firm's return with dividends and without dividends (*ret* and *retx* from CRSP, respectively) is positive. Tobin's q is book value plus market equity minus book equity, divided by book value. Return on assets, ROA, is EBITDA divided by total assets. Special meeting is a binary variable equal to 1 for special meetings. Institutional ownership is equal to the number of shares owned by institutions divided by the shares outstanding in the year prior to the meeting, both from Thomson Reuters. ISS against management is a binary variable that equals 1 if ISS has a recommendation other than "For" for a management proposal, or a "For" recommendation for a shareholder proposal. Log account value is the log of the total account value for that account in the calendar year defined as the sum across all firms held by the account of the product of share price and the number of shares owned. 2016 county presidential turnout is the number of county residents who cast ballots in the 2016 U.S. presidential election from CQ Voting and Elections, divided by the number of adult citizens from the Census Bureau. Log zip code AGI is the average adjusted gross income in the prior calendar year in the account's zip code. Fraction over 65 is the fraction of zip-code residents above the age 65, from the Census, defined as $(DPSF0010015 + DPSF0010016 + DPSF0010017 + DPSF0010018 + DPSF0010019)/DPSF0010001$. Density is the population divided by land area in square meters ($DPSF0010001/AREALAND$). Fraction with bachelors and fraction with post-bachelors are zip-code-level five-year averages from the U.S. Census as of 2017. Fraction in Finance/Insurance is equal to the number of employed workers in Finance/Insurance divided by all-industries employment, both at the zip-code level, from the Bureau of Labor Statistics. Column (1) includes proposal category, industry, and year-month fixed effects; column (2) includes proposal fixed effects; column (3) includes proposal category, industry, and account-year fixed effects; column (4) includes account-meeting and account-proposal category fixed effects; column (5) includes proposal and account-year fixed effects; and column (6) includes proposal, account-year, and account-firm fixed effects. Industry fixed effects use Fama French industry categories; proposal-category fixed effects use the proposal categories set forth in Online Appendix Table A1. All right-hand-side variables are demeaned, so that the intercept reflects the turnout of an account with average levels of each covariate. Observations are weighted by the inverse of the number of meetings for the account-year, so that each account-year is weighted equally. Standard errors clustered at the account and meeting level are in parentheses. Number of clusters refers to the number of distinct meetings. *, **, and *** represent significance at the 0.05, 0.01, and 0.001 levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
Log(α)	1.098***	0.823***	0.458***		0.238***	−0.350
	(0.061)	(0.064)	(0.044)		(0.045)	(0.285)
Log(ME)	1.036***		0.605***			
	(0.116)		(0.063)			
Yearly abnormal return	4.404***		2.469***			
	(0.705)		(0.269)			
Dividend indicator	1.370*		0.107			
	(0.560)		(0.252)			
Tobin's q	0.334		0.130			
	(0.191)		(0.091)			
ROA	7.027**		3.174**			
	(2.690)		(1.229)			
Special meeting	−4.709***		−2.865***			
	(1.143)		(0.763)			
Institutional ownership	−1.790		0.836			
	(1.058)		(0.489)			
ISS against management	−2.725***		−2.429***	−1.483***		
	(0.483)		(0.373)	(0.311)		
Log account value	0.020	0.061**				
	(0.023)	(0.022)				
2016 county presidential turnout	−1.436	−1.208				
	(1.118)	(1.078)				
Log zip code income	1.934***	1.884***				
	(0.300)	(0.292)				
Fraction over 65	6.005***	5.825***				
	(1.120)	(1.101)				
Density	−0.000***	−0.000***				
	(0.000)	(0.000)				
Fraction with bachelors	−3.491*	−0.020				
	(1.553)	(0.015)				
Fraction with post-bachelors	−7.065**	−0.092***				
	(2.563)	(2.54)				
Fraction in Finance/Insurance	3.413	5.893				
	(7.613)	(7.295)				
Intercept	85.495***	85.690***	86.357***	87.980***	86.574***	86.559***
	(0.187)	(0.117)	(0.046)	(0.008)	(0.008)	(0.055)
Proposal Category FE	Yes		Yes			
Industry FE	Yes		Yes			
Year-Month FE	Yes					
Proposal FE		Yes			Yes	Yes
Account-Year FE			Yes			
Account-Meeting FE				Yes		

(continued on next page)

Table 9

(*continued*)

	(1)	(2)	(3)	(4)	(5)	(6)
Account-Proposal Category FE				Yes		
Account-Year FE					Yes	Yes
Account-Firm FE						Yes
N	7,388,040	7,488,217	7,771,765	7,701,840	7,880,494	7,856,887
Number of clusters	7,239	6,794	7,591	5,056	7,460	6,772
R^2	0.09	0.15	0.58	0.81	0.60	0.65

5.9 Table 10 and Figure 5: Retail versus institutional investors and ISS sensitivity

Read: Institutions respond much more strongly to ISS opposition than retail shareholders. This difference persists across account value, turnover, and breadth bins.

Detailed interpretation: The result points to structural differences between retail and institutional voting, not merely differences in size or sophistication.

Economic message: Retail shareholders would not replicate institutional stewardship if given more voting power.

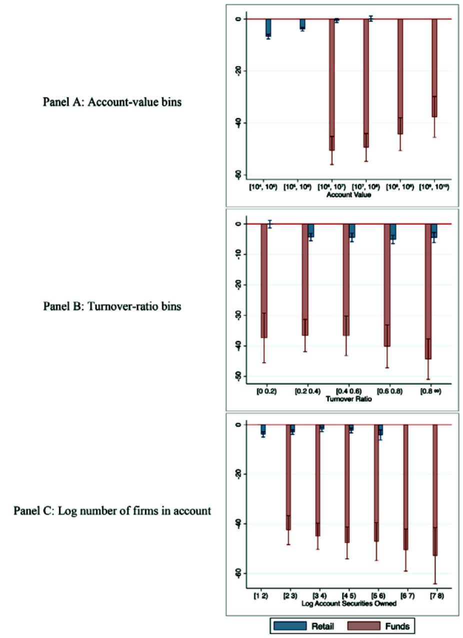
Table 10

Comparison of retail and institutional investors' decisions.

This table reports regression results on shareholder voting with votes aggregated to the proposal level. The dependent variable is the number of votes cast in line with management's recommendation divided by the number of votes cast For or Against, multiplied by 100. Log market equity is the log of market equity computed as price time shares outstanding from CRSP, as of the record date month. Yearly abnormal return is the firm buy-and-hold return for the period 13 months to 1 month prior to the record date, minus the buy-and-hold value-weighted market return from CRSP. The dividend indicator is a binary variable equal to 1 if the difference in the firm's buy-and-hold return with dividends and without dividends (*ret* and *retx* from CRSP, respectively) is positive. Tobin's q is book value plus market equity minus book equity, divided by book value. Return on assets, ROA, is EBITDA divided by total assets. Special meeting is a binary variable equal to 1 for special meetings. Institutional ownership is equal to the number of shares owned by institutions divided by the shares outstanding in the year prior to the meeting, both from Thomson Reuters. ISS against management is a binary variable that equals 1 if ISS has a recommendation other than For for a management proposal, or a For recommendation for a shareholder proposal. Columns (1) through (3) include institutional voting results and columns (4) through (6) contain retail shareholder voting results. All columns except (3) and (6) include industry fixed effects, and columns (3) and (6) include firm fixed effects. Industry fixed effects use Fama French industry categories; time fixed effects are at the year-month level; proposal-category fixed effects use the proposal categories set forth in Online Appendix Table A1. All right-hand-side variables are demeaned over all observations in the sample, so the intercept reflects the average vote for an observation with mean values of those covariates. Observations are weighted so that each meeting is weighted equally. Standard errors clustered at the meeting level are in parentheses. Number of clusters refers to the number of distinct meetings. *, **, and *** represent significance at the 0.05, 0.01, and 0.001 levels, respectively.

	Institutional voters			Retail voters		
Vote with management	(1)	(2)	(3)	(4)	(5)	(6)
Log(ME)	0.824***	0.751***		0.527***	0.527***	
	(0.148)	(0.100)		(0.086)	(0.086)	
Yearly abnormal return	0.109	−1.105*	−0.413	4.369***	4.396***	2.853***
	(0.764)	(0.503)	(0.439)	(0.424)	(0.423)	(0.329)
Dividend indicator	2.096***	−0.396	−3.555**	−0.326	−0.437	1.708*
	(0.496)	(0.316)	(1.085)	(0.284)	(0.283)	(0.788)
Tobin's q	0.282	0.387**	0.609	0.332***	0.331***	0.491*
	(0.181)	(0.119)	(0.321)	(0.098)	(0.098)	(0.203)
Return on assets	7.477***	1.729	−1.382	3.842***	3.734***	1.442
	(1.799)	(1.155)	(2.148)	(0.876)	(0.883)	(1.312)
Special meeting	−7.769***	−3.603***	−3.032**	−1.118	−1.000	−0.234
	(1.482)	(0.904)	(1.025)	(0.739)	(0.742)	(0.656)
Institutional ownership	6.760***	4.074***	3.562	2.743***	2.703***	−0.519
	(1.037)	(0.679)	(2.200)	(0.581)	(0.583)	(1.746)
ISS against management		−50.721***	−46.684***		−1.781***	−1.802***
		(0.787)	(0.709)		(0.428)	(0.330)
Intercept	88.335***	88.449***	88.780***	89.334***	89.305***	89.570***
	(0.230)	(0.149)	(0.148)	(0.127)	(0.127)	(0.106)
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes		Yes	Yes	No
Proposal Category FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE			Yes			Yes
N	33,116	32,998	32,674	33,392	33,263	32,942
Number of clusters	7,781	7,771	7,447	7,884	7,873	7,552
R^2	0.14	0.62	0.77	0.17	0.17	0.65

(a) Table 10



voting to ISS recommendations.

(b) Figure 5

5.10 Table 11: Retail shareholder voting decisions and exit

Read: Voting with management predicts continued ownership; disagreement with management predicts exit.

Detailed interpretation: This connects voice and exit: ballots reveal investor attitudes that also affect subsequent portfolio decisions.

Economic message: Retail votes are related to real ownership decisions, not just symbolic ballot choices.

Table 11
Retail shareholder voting decisions and exit.
This table describes the relation between changes in ownership between years t and $t+1$ and firm and account characteristics. The data are limited to accounts that, at year t, hold firms that appear in the data in year $t+1$. The dependent variable, still own next year, is equal to 1 if the account holds the firm in $t+1$. Cast ballot is an indicator variable equal 1 if the account cast a ballot in t. α is defined as the account's number of shares divided by the firm's number of shares outstanding as of the record date month, from CRSP. Log market equity is the log of market equity computed as price time shares outstanding from CRSP, as of the record date month. Yearly abnormal return refers to the firm's buy-and-hold return for the period 13 months to 1 month prior to the record date minus the value-weighted market return from CRSP. The dividend indicator is a binary variable equal to 1 if the difference in the firm's return with dividends and without dividends (*ret* and *retx* from CRSP, respectively) is positive. Tobin's q is book value plus market equity minus book equity, divided by book value. ROA, return on assets, is EBITDA divided by total assets. Institutional ownership is equal to the number of shares owned by institutions divided by the shares outstanding in the year prior to the meeting, both from Thomson Reuters. Yearly abnormal return, the dividend indicator, return on assets, and Tobin's q are as of year $t+1$. In columns (2)–(4), we start from a random sample of accounts and limit the analysis to account meetings in which the account cast a ballot. WithMGMT is the fraction of proposals at year t on which the account voted in line with management. We also include WithMGMT for the following subcategories of proposals: (i) management-sponsored proposals; (ii) shareholder-sponsored proposals, and certain subcategories of management-sponsored proposals; (iii) director proposals; (iv) say-on-pay proposals; and (v) other management-sponsored proposals. All columns include year-month, industry, and account-year fixed effects. Industry fixed effects use Fama French industry categories. Proposal category fixed effects use the proposal categories in Online Appendix Table A1. Observations are weighted by the inverse of the number of meetings for the account-year, so that each account-year is weighted equally. Standard errors clustered at the account and meeting level are in parentheses. Number of clusters refers to the number of distinct meetings. *, **, and *** represent significance at the 0.05, 0.01, and 0.001 levels, respectively.

	Unconditional	Conditional on turnout		
Still own next year	(1)	(2)	(3)	(4)
Cast ballot$_t$	1.112***			
	(0.153)			
WithMGMT{on all proposals}		1.511***		
		(0.205)		
WithMGMT{on management proposals}			1.466***	
			(0.282)	
WithMGMT{on shareholder proposals}			0.451***	0.473*
			(0.108)	(0.233)
WithMGMT{on director proposals}				3.306***
				(0.535)
WithMGMT{on say-on-pay proposals}				0.716**
				(0.264)
WithMGMT{on other management proposals}				0.006
				(0.279)
Log(α_t)	0.826***	1.388***	1.493***	1.557***
	(0.063)	(0.066)	(0.072)	(0.120)
Log(ME$_t$)	1.662***	2.165***	2.657***	3.024***
	(0.122)	(0.105)	(0.212)	(0.496)
Institutional ownership$_t$	−4.558***	1.222	3.608*	3.648*
	(0.874)	(0.661)	(1.395)	(1.548)
Yearly abnormal return$_{t+1}$	−0.113	0.744	−0.969	1.091
	(0.555)	(0.490)	(0.879)	(1.125)
Dividend indicator$_{t+1}$	−0.174	0.661***	0.404*	1.076**
	(0.582)	(0.121)	(0.198)	(0.366)
Tobin's q$_{t+1}$	0.799***	0.192	1.065	−10.371*
	(0.150)	(0.807)	(2.121)	(4.778)
ROA$_{t+1}$	−0.206	−4.114***	−3.392*	−1.135
	(0.788)	(0.735)	(1.413)	(3.429)
Intercept	69.848***	41.001***	32.002***	17.895

Table 11

6 Result map

- Retail ownership is economically large but dispersed.
- Retail turnout is low but systematic.
- Retail votes can affect outcomes in close contests.
- Voting access and information-material delivery methods matter for turnout.
- Retail voters monitor management and react to firm performance.
- Retail voting differs from institutional voting and is less ISS-sensitive.
- Voice and exit are linked: disagreement with management predicts future selling.

7 Short summary

The paper shows that retail shareholders are individually small but collectively meaningful. Their turnout responds to ownership, benefits, and costs. Their votes can swing outcomes in close contests. Conditional on voting, they monitor management and differ from institutional investors.

8 Conclusion

8.1 Core conclusion

Retail shareholders are not merely passive owners. They vote selectively, react to incentives, and provide a distinct form of voice.

8.2 Economic intuition

Retail voting combines instrumental benefits, costs of participation, and expressive benefits. The collective-action problem is severe but not complete.

8.3 Policy implications

Proxy-plumbing reforms should reduce voting costs without removing channels used by non-internet or paper-reliant shareholders. Pass-through voting reforms should not assume that retail shareholders mimic institutional votes.

8.4 Limitations

The anonymized data do not identify all household characteristics or holdings across brokers, and many voting-choice specifications remain observational.

8.5 Takeaway

Retail voice is real, distinct, and sensitive to institutional design.